

Personal Credit Risk Assessment Based on Swarm Intelligence and Random Forest

Hanyin Zhang¹, Facheng Dai², Xiangguang Dai^{2*}

¹Washington University in St. Louis

²Chongqing Three Gorges University

* daixiangguang@163.com

Abstract—Personal credit risk assessment plays a critical role in the issuance of credit loans by banks and other financial institutions, fundamentally serving as a predictive problem. However, traditional methods often only achieve local optimal solutions. This paper introduces an innovative method for personal credit risk assessment by integrating swarm intelligence and random forest techniques, aiming to find the global optimal solution. The practicality and efficiency of this algorithm have been demonstrated through experimental validation on the German credit database.

Index Terms—Credit assessment, PSO, Random Forest, Prediction

I. INTRODUCTION

With the rapid economic growth, the credit services industry has also achieved significant progress. However, as the personal financial services of commercial banks rapidly expand, the risk of credit default is also increasing. The major issues in the lending industry include the lack of a fully developed credit system in the commercial sector, the severe absence of credit records for market participants, the scarcity of incentives for maintaining trust, coupled with the low cost of dishonor, the underdevelopment of the credit service market and the insufficient credibility of its institutions, the pending establishment of a protection mechanism for the interests of credit information subjects, the low level of credit awareness and credibility among market participants, the incomplete adoption of honesty and trustworthiness as a commercial norm in the market, and the persistent occurrence of dishonest practices such as business fraud, fake product manufacturing and sales, tax evasion, and other illegal activities [1], [2]. Relying solely on the personal experience of credit officers to assess the creditworthiness of borrowers poses significant risks.

As the banking loan business evolved and attained maturity, leading banks started leveraging a statistical credit scoring mechanism to forecast borrowers' credit risks [3]. This mechanism employed a linear model, incorporating factors like age, marital status, salary, asset profile, and annual company sales as inputs. Through a series of computations, it generated a credit risk index for each borrower. Credit officers referred to this index in making credit decisions, often denying applications if the risk assessment was excessively high.

While this scoring system effectively mitigated credit risks to some extent, it gradually proved inadequate in addressing

the complexities of real-world scenarios. Borrowers' characteristics often exhibited intricate nonlinear relationships and mutual dependencies, making traditional methods less effective in assessing credit risks [4]. In the recent past, the advent of cutting-edge computer technologies like big data and artificial intelligence has revolutionized bank credit risk prediction. These modern information technologies have significantly reduced credit risks associated with information asymmetries and imperfections [5]. Deep learning, which has emerged in recent years, has made leapfrog progress in areas such as image, video, speech, and natural language processing. How to integrate these advanced machine learning methods with individual credit risk assessment in the context of big data in the financial sector is a worthwhile research direction.

II. RELATED WORKS

In 1980, Wiginton pioneered the application of logistic regression in loan credit evaluation, yielding impressive results [6]. By 1985, Makowski took a leap forward with a decision tree-based model for predicting loan credit risks, significantly boosting predictive accuracy [7]. These early models, rooted in rigorous mathematical theories, reduced loan risks for banks and offered a degree of predictive certainty, yet their accuracy still held vast potential for improvement.

Capotorti and colleagues then introduced a novel approach to loan credit risk assessment, leveraging rough sets [8]. Bekhe and his team further refined the field by integrating Logistic regression and radial basis function models, creating a feature-fusion-based method that excelled in accuracy [9]. Meanwhile, Oreski et al. explored genetic algorithms for credit risk assessment, offering a unique perspective [10]. Boughaci D and his colleagues merged previous models, not only enhancing prediction accuracy but also enabling feature selection during the modeling process [11]. Maldonado et al. capitalized on support vector machines for feature selection in credit risk models, ensuring the selection of relevant features [12]. Sang H V and his team quantified the feature selection process with a mathematical scoring method, further refining the credit risk assessment process [13].

Thomas innovatively used Markov chains to establish transition probability matrices, leading to the development of scoring models that revolutionized credit risk assessment [14]. Blanco et al. broke new ground with a credit risk assessment model for microloans, leveraging shallow neural networks

[15]. Ignatius et al. employed the analytic hierarchy process to calculate feature weights, providing a more nuanced approach to credit risk calculation [16].

After 2015, the rapid evolution of deep learning technologies opened new horizons in loan credit risk prediction. Angelini et al. spearheaded the use of conventional deep neural networks in credit risk modeling, marking a significant milestone [17]. Zhou et al. merged unsupervised and supervised learning, creating a combined loan credit risk assessment model that revolutionized client scoring [18]. Bequé et al. designed a deep neural network specifically for loan credit risk assessment, refining the convergence methods of traditional neural networks [19]. Simulation experiments confirmed the network's superiority in terms of fewer parameters and high accuracy.

This article aims to thoroughly analyze and compare the practical effects of traditional risk assessment methods and deep learning techniques in personal credit risk decision-making, in order to reveal the strengths and limitations of each approach.

III. RELATED ALGORITHMS

A. Decision Tree

The core concept of a decision tree lies in its hierarchical structure, which mimics the decision-making process in human thinking. Each internal node in the tree signifies a test conducted on a specific attribute, whereas each branch depicts a potential outcome resulting from that test. Conversely, the leaf nodes signify categorical labels or conclusive decisions. This architecture facilitates a gradual and methodical evaluation of the data, ultimately leading to a conclusive decision or prediction.

In the context of personal credit risk assessment, decision trees can play a significant role. By using relevant features such as income, employment history, and credit history, a decision tree can effectively classify individuals into different risk categories. In turn, this can aid financial institutions in making lending decisions that are more informed, thereby mitigating the risk of default and enhancing the overall profitability of their credit portfolios.

During the construction of decision trees, the method of information gain is frequently employed to establish the order of data attributes. Given that the proportion of samples belonging to the k -th class (among a total of n classes) within the current sample set D is p_k , the information entropy of sample set D is computed utilizing equation 1. A lower value of $Ent(D)$ indicates a higher degree of purity within the sample class.

$$Ent(D) = - \sum_{k=1}^n p_k \log_2(p_k) \quad (1)$$

For each attribute of the data, calculate the information gain $Gain(D, a)$ obtained by dividing attribute a into sample set D by equation 2. The attribute with higher information gain will be ranked closer to the root node in the decision tree.

$$Gain(D, a) = Ent(D) - \sum_{v=1}^V \frac{|D^v|}{|D|} Ent(D^v) \quad (2)$$

B. Random Forest

Random forests are a robust and effective machine learning algorithm that offer improved predictive performance compared to individual decision trees. Their ability to handle diverse data types, robustness to noise, and interpretability make them a popular choice in various applications, including credit risk assessment. With continued advancements in machine learning, random forests are likely to remain a key component in the toolbox of data scientists and analysts.

The essence of a random forest is a decision tree. Assuming the number of training sets is N , for each tree in it, randomly and with replacement, select training samples from the original training set, repeat K times, and obtain K sets of training samples. If the sample dimension of each feature is M , randomly select m features from the M features, and note that m is much smaller than M . Train each tree using these m features to obtain K classification models, and through the final vote, select the optimal result with the highest probability.

C. Support Vector Machine

Support Vector Machine (SVM) is a supervised machine learning algorithm that finds widespread application in handling classification problems, particularly in binary classification scenarios, given the training set $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$ and $y_i \in \{-1, 1\}$, the objective of the Support Vector Machine is to identify a hyperplane that effectively separates samples of different categories and possesses superior generalization capabilities. In the sample space, this hyperplane is mathematically represented by an equation 3:

$$w^T x + b = 0 \quad (3)$$

The equation of the hyperplane is defined by a normal vector $w^T = \{w_1, w_2, \dots, w_d\}$ and a displacement term b . The distance from any point x in the sample space to this hyperplane is represented by:

$$r = \frac{||w^T x + b||}{||w||} \quad (4)$$

If the hyperplane is able to accurately classify the training data, the following results exist for any (x_i, y_i) within D :

$$\begin{cases} w^T x_i + b \geq +1, y_i = +1 \\ w^T x_i + b \leq -1, y_i = -1 \end{cases} \quad (5)$$

Integrating equation 5 yields equation 6:

$$y_i \cdot (w^T x_i + b) \geq 1 \quad (6)$$

The points that lie on either side of the hyperplane and are closest to it are referred to as support vectors. The sum

of the distances between these two support vectors and the hyperplane is calculated as:

$$r = \frac{2}{\|w\|} \quad (7)$$

At this point, the problem is transformed into finding a hyperplane that maximizes r . Find the parameters w and b to maximize r . The Lagrange multiplier method can be used to calculate the parameters w and b :

$$L = \frac{1}{2}\|w\|^2 + \sum_{i=1}^m \alpha_i(1 - y_i(w^T x_i + b)) \quad (8)$$

Then, on equation 9, take the partial derivatives of w and b separately, and set these two partial derivatives to zero, as follows:

$$\begin{aligned} \frac{\partial L}{\partial w} = w - \sum_{i=1}^m \alpha_i y_i x_i = 0 \rightarrow w = \sum_{i=1}^m \alpha_i y_i x_i \\ \frac{\partial L}{\partial b} = \sum_{i=1}^m \alpha_i y_i \end{aligned} \quad (9)$$

After substituting these two partial derivatives into equation 8 and performing the operation, we have:

$$L = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j x_i^T x_j \quad (10)$$

Finally, by solving equation 10, the vector can be obtained α , w and b .

D. Multi-layer perceptron

Multi-Layer Perceptron (MLP) constitute a fundamental neural network type, serving as the cornerstone for numerous deep learning architectures. These networks, composed of numerous interconnected layers of neurons, excel at learning intricate representations and patterns from input data. Consequently, they are highly effective for a diverse array of tasks, encompassing classification, regression, and function approximation.

The essence of an MLP's structure lies in its composition: an input layer, one or multiple hidden layers, and an output layer. Each layer comprises a set of neurons, where each neuron receives inputs from the preceding layer. It then applies a weighted sum to these inputs and passes the resulting value through a nonlinear activation function, generating an output. This mechanism enables the network to capture and represent nonlinear relationships inherent in the data.

Training an MLP involves meticulously adjusting the weights and biases of its neurons to minimize a loss function. This function quantifies the disparity between the network's predictions and the actual targets. Typically, gradient descent or its variants, like stochastic gradient descent and mini-batch gradient descent, are employed to achieve this optimization. The gradients of the loss function with respect to the weights and biases are computed using the chain rule of calculus, a process known as backpropagation.

IV. ALGORITHM DESIGN

Random Forest, as an ensemble learning method, significantly enhances prediction accuracy and stability by aggregating the predictions of multiple decision trees. However, the inherent randomness in feature and sample selection during its construction can sometimes limit its performance.

Particle Swarm Optimization (PSO) is a swarm intelligence-based optimization algorithm that mimics the foraging behavior of bird flocks to efficiently search for optimal solutions in a given search space. When applied to improve Random Forest, PSO can be utilized for both feature selection and parameter optimization.

Regarding feature selection, the traditional Random Forest approach randomly selects features for each tree, potentially leading to the exclusion of crucial features and affecting model performance. To address this, we harness the global search capabilities of PSO to identify the optimal feature subset within the feature space. In this approach, each particle represents a potential feature combination, and its position is iteratively updated based on its own experience and the knowledge of the swarm. This process gradually guides the particles towards the optimal feature combination, enabling the retention of informative features while eliminating redundant and noisy ones, thereby enhancing the prediction accuracy of Random Forest.

For parameter optimization, Random Forest's performance is influenced by various parameters, such as the number of trees and the maximum depth of each tree. The selection of these parameters is crucial for achieving optimal model performance. Traditional methods like grid search or random search can be computationally inefficient and prone to getting stuck in local optima. To overcome these limitations, we integrate PSO into the parameter tuning process of Random Forest. By encoding parameter configurations as particle positions and allowing particles to explore the parameter space through iterative updates, PSO can efficiently identify the optimal parameter combination that maximizes model performance.

By combining the strengths of Random Forest and PSO, we can create a more robust and accurate predictive model that takes advantage of the ensemble learning capabilities of Random Forest while overcoming its limitations through intelligent feature selection and parameter optimization using PSO.

PSO initializes a group of particles, which represent random solutions, and subsequently searches iteratively for the optimal solution. Within each iteration, particles refine themselves by monitoring two key values: $pbest_i$, which stands for their personal best solution, and $gbest$, representing the global best solution found so far. Once these two optimal values are identified, particles adjust their velocity and position using following equation:

$$\begin{aligned} v_i = v_i + c_1 \cdot rand() \cdot (pbest_i - x_i) \\ + c_2 \cdot rand() \cdot (gbest - x_i) \end{aligned} \quad (11)$$

$$x_i = x_i + v_i \quad (12)$$

i represents the index of the particles, with $i = 1, 2, \dots, N$. N denotes the total number of particles in the swarm and v_i denotes the velocity of the particles. The maximum allowed value for v_i is set as $V_{max} > 0$. If $v_i > V_{max}$, then $v_i = V_{max}$. $Rand()$ is a random number between $(0, 1)$. The current position of the x_i particle; c_1 and c_2 are learning factors, typically $c_1 = c_2 = 2$.

In the equation referenced as 11, three distinct components play a crucial role. The first component, known as the memory term, encapsulates the impact of the previous velocity's magnitude and direction. This term preserves a sense of inertia, allowing particles to maintain some momentum from their prior movements.

The second component, labeled the self-cognition term, represents a vector directed from the current position to the particle's best-known position. This vector underscores the influence of the particle's own experiences, drawing it towards previously successful locations.

The third component, termed the social-cognition term, is a vector pointing from the current position to the best-known position within the entire population. This term reflects the collaborative nature of PSO, where particles learn from each other's successes, fostering knowledge sharing and cooperation.

Based on the above two equation, the standard form of PSO is established.

$$v_i = \omega \cdot v_i + c_1 \cdot rand() \cdot (pbest_i - x_i) + c_2 \cdot rand() \cdot (gbest_i - x_i) \quad (13)$$

where ω is referred to as the inertia factor, and its value is non-negative. When ω has a larger value, the global optimization ability is strong, while the local optimization ability is weak. Conversely, when ω has a smaller value, the global optimization ability is weak, while the local optimization ability is strong. Dynamic ω can achieve better optimization results than a fixed value. Dynamic ω can vary linearly during the PSO search process, or it can dynamically change based on a certain measurement function of PSO performance.

Currently, the most commonly adopted strategy is the linearly decreasing weight strategy:

$$\omega^{(t)} = (\omega_{ini} - \omega_{end})(G_k - g)/G_k + \omega_{end} \quad (14)$$

In this strategy, G_k represents the maximum number of iterations; ω_{ini} is the initial inertia weight; ω_{end} is the inertia weight when iterating to the maximum evolution generation; and a typical weight value is $\omega_{ini} = 0.9, \omega_{end} = 0.4$.

The introduction of ω has greatly improved the performance of the PSO algorithm. By adjusting the global and local search capabilities, it can be tailored to different search problems, making the PSO algorithm successfully applicable to many practical issues. The detailed algorithm is provided in the pseudo-code of Algorithm 1.

Algorithm 1 PSO-RF

Input: $\{a, b, c, \dots\}_n$ (A total of n borrower information)

Output: $\{X\}_n$ (Binary Classification Result)

- 1: Generate a swarm Ω , Initialize $\{a, b, c, \dots\}_n, \{X_i\}_n$ and set their velocities as zeros, where $i = 1, \dots, N$.
 - 2: $seed_i$ (Random Seed)
 - 3: Initialize precision(PRE), recall(REC), F-Score
 - 4: **for** $i = 1$ to N **do**
 - 5: $(\{X\}_n) = RF(\{a, b, c, \dots\}_n, seed_i)$
 - 6: Select the current best positions $\{X_i\}_n, seed_i$ by calculate PRE, REC, F-Score.
 - 7: Reset $\{X_i\}_n, seed_i$ and their velocities by equation 13
 - 8: **end for**
 - 9: **return** $\{X\}_n$
-

V. EXPERIMENTS

The assessment methods for personal credit primarily consist of qualitative and quantitative assessments. The former relies mainly on the subjective judgment of credit personnel, while the latter analyzes personal customer data using tools such as scorecards and credit scoring models. Qualitative data in the dataset is processed through one-hot encoding.

The data used in this paper is sourced from the German credit database, which was organized and compiled by Professor Hans Hofmann. It contains 1000 customer profiles, with each customer having 20 attributes. Additionally, each profile is labeled with either a good or bad credit rating.

This article utilizes decision tree, random forest, support vector machine, and multi-layer perceptron, to train and predict the given dataset. The effectiveness of these approaches is then assessed through comprehensive evaluation metrics, including precision(PRE), recall(REC), and F-Score, to comprehensively understand their respective performances.

TABLE I
PRF, REC AND F-SCORE

	PRE	REC	F-Score
DT	0.73988	0.73988	0.73988
RF	0.90173	0.76197	0.82539
SVM	0.87861	0.78350	0.82833
MLP	0.79213	0.81502	0.80341
PSO-RF	0.92134	0.82304	0.85698

According to the results shown in the table I, the PSO-RF achieved the best performance in personal credit risk prediction. The integration of PSO further improves the performance of Random Forest by optimizing the parameters of the ensemble, such as the number of trees or the depth of each tree.

By leveraging the strengths of both Random Forest and PSO, the PSO-RF algorithm is able to achieve superior prediction accuracy in personal credit risk assessment. It can effectively handle high-dimensional and nonlinear data, while

being resilient to noise and outliers. Additionally, the dynamic adjustment of parameters through PSO allows the algorithm to adapt to different datasets and scenarios, further enhancing its generalization capabilities.

In contrast, the random forest without PSO algorithm improvement also showed some predictive ability, but the effect was slightly inferior to the former. This may be due to the lack of parameter optimization in the random forest, which may not achieve the optimal balance between model complexity and generalization ability.

Multilayer Perceptron, as a deep learning model, exhibits slightly inferior performance compared to Random Forest. This may be due to MLP's greater sensitivity to factors such as data distribution, initial weight initialization, and learning rates during training, leading to reduced generalization capabilities compared to Random Forest. Nevertheless, MLP retains advantages in handling high-dimensional and nonlinear features, resulting in satisfactory prediction performance in certain scenarios.

Support Vector Machine demonstrates moderate performance in predicting personal credit risk. This could be attributed to SVM's sensitivity to the scale and distribution of data. In the context of credit risk prediction, the imbalance in data, specifically the disproportionate ratio of defaulting and non-defaulting customers, may affect SVM's performance. Additionally, SVM may encounter computational complexity and performance degradation when dealing with high-dimensional data.

Decision Trees exhibit the worst performance, potentially due to their susceptibility to overfitting, especially in the presence of noisy data or strong feature correlations. Furthermore, Decision Trees employ hard partitioning in dividing the feature space, which may not effectively capture complex relationships within the data. While Decision Trees are easy to understand and implement, their performance may fall short when dealing with complex problems compared to other more sophisticated models.

VI. CONCLUSION

The improvement of the random forest model by the PSO algorithm significantly enhances prediction accuracy. This is mainly due to the powerful parameter optimization capabilities of the PSO algorithm, which precisely adjusts the parameters of the random forest to better adapt to complex data distributions and feature relationships. This optimization not only improves the prediction performance of the model but also enhances its stability and generalization ability, enabling it to maintain high accuracy when facing new credit data.

Compared with other algorithms, the PSO-RF algorithm demonstrates clear advantages. Although the random forest without PSO algorithm improvement exhibits certain predictive ability, its performance is not optimal due to the lack of parameter optimization. As a neural network method, the multilayer perceptron (MLP) has advantages in dealing with nonlinear problems, but in this case, due to data characteristics and task complexity, its effects did not exceed the PSO-RF

algorithm. Although support vector machines (SVM) usually perform well in handling high-dimensional data and classification problems, in the credit risk prediction task of this experiment, their performance was inferior to the PSO-RF algorithm. Decision trees are prone to overfitting and have poor generalization ability when dealing with complex problems, resulting in the worst performance in this case.

We can conclude that the PSO-RF algorithm has achieved excellent results in personal credit risk prediction. This conclusion not only provides a new and effective tool for the field of credit risk management but also offers a useful reference for solving other similar problems. In the future, we can further explore the application of the PSO-RF algorithm in other fields and continuously optimize its performance to adapt to more complex data and task requirements.

REFERENCES

- [1] Xu, D. (2017). Suggestions on innovation development of commercial credit system. *Macroeconomic Management*, 2, 62-66.
- [2] Xu, D., & Ma, J. (2018). The Credit Asset of Enterprise Accounts Receivable Pricing Model. *Complexity*, 2018.
- [3] Chen, S. F., Charkaborty, G., Li, L. H., & Lin, C. T. (2019). Credit risk assessment using regression model on P2P lending. *International Journal of Applied Science and Engineering*, 16(2), 149-157.
- [4] Soui, M., Gasmi, I., Smiti, S., & Ghédira, K. (2019). Rule-based credit risk assessment model using multi-objective evolutionary algorithms. *Expert Systems with Applications*, 126, 144-157.
- [5] Lusto, R. (2019). Credit Risk Assessment for Motorcycle Loan Application: A Fuzzy Logic Approach [J]. *Journal of Advanced Research in Dynamical and Control Systems*, 11, 1246-1252.
- [6] Wiginton, J. C. (1980). A note on the comparison of logit and discriminant models of consumer credit behavior. *Journal of Financial and Quantitative Analysis*, 15(3), 757-770.
- [7] Makowski, P. (1985). Credit scoring branches out. *Credit World*, 75(1), 30-37.
- [8] Capotorti, A., & Barbanera, E. (2012). Credit scoring analysis using a fuzzy probabilistic rough set model. *Computational Statistics & Data Analysis*, 56(4), 981-994.
- [9] Bekhet, H. A., & Eletter, S. F. K. (2014). Credit risk assessment model for Jordanian commercial banks: Neural scoring approach. *Review of Development Finance*, 4(1), 20-28.
- [10] Oreski, S., & Oreski, G. (2014). Genetic algorithm-based heuristic for feature selection in credit risk assessment. *Expert systems with applications*, 41(4), 2052-2064.
- [11] Boughaci, D., & Alkhawaldeh, A. A. S. (2018). Three local search-based methods for feature selection in credit scoring. *Vietnam Journal of Computer Science*, 5, 107-121.
- [12] Maldonado, S., Bravo, C., López, J., & Pérez, J. (2017). Integrated framework for profit-based feature selection and SVM classification in credit scoring. *Decision Support Systems*, 104, 113-121.
- [13] Ha, V. S., & Nguyen, H. N. (2016). FRFE: Fast recursive feature elimination for credit scoring. In *Nature of Computation and Communication: Second International Conference, ICTCC 2016, Rach Gia, Vietnam, March 17-18, 2016, Revised Selected Papers 2* (pp. 133-142). Springer International Publishing.
- [14] Thomas, L. C. (2000). A survey of credit and behavioural scoring: forecasting financial risk of lending to consumers. *International journal of forecasting*, 16(2), 149-172.
- [15] Blanco, A., Pino-Mejías, R., Lara, J., & Rayo, S. (2013). Credit scoring models for the microfinance industry using neural networks: Evidence from Peru. *Expert Systems with applications*, 40(1), 356-364.
- [16] Ignatius, J., Hatami-Marbini, A., Rahman, A., Dhamotharan, L., & Khoshnevis, P. (2018). A fuzzy decision support system for credit scoring. *Neural Computing and Applications*, 29, 921-937.
- [17] Angelini, E., Di Tollo, G., & Roli, A. (2008). A neural network approach for credit risk evaluation. *The quarterly review of economics and finance*, 48(4), 733-755.

- [18] Huang, D., Zhou, J., & Wang, H. (2018). RFMS method for credit scoring based on bank card transaction data. *Statistica Sinica*, 28(4), 2903-2919.
- [19] Bequé, A., & Lessmann, S. (2017). Extreme learning machines for credit scoring: An empirical evaluation. *Expert Systems with Applications*, 86, 42-53.